

SERAT M. SAAD

Ph.D. Student in Astronomy, The Ohio State University

saad.104@osu.edu | seratsaad.github.io | ORCID 0009-0004-9592-2311

RESEARCH INTERESTS

Dynamics: binary stars, radial migration, tests of Modified Newtonian Dynamics (MOND), an alternative to dark matter

Stellar Spectroscopy: chemical abundances, metal-rich stars, star spots, young stellar objects

AI for Astronomy: LLM agents, agent coordination, automation of astrophysical research

EDUCATION

Ph.D. in Astrophysics (in progress), The Ohio State University <i>Advisors: Prof. Yuan-Sen Ting and Prof. Krzysztof (Kris) Stanek</i>	2025-Present
A.B. in Physics & Applied Mathematics, Vanderbilt University <i>Thesis: Understanding Accretion Activity in Young Stars using SDSS-V BOSS Spectra</i> <i>Advisor: Prof. Keivan Stassun</i>	2021-2025

HONORS & AWARDS

University Distinguished Fellowship, The Ohio State University <i>graduate-school fellowship supporting doctoral study</i>	2029-2030
University Fellowship, The Ohio State University <i>graduate-school fellowship supporting doctoral study</i>	2025-2027
Hult Prize, National Finalist <i>for a student venture expanding access to personalized education</i>	2026
Larry R. Cathey Award, Vanderbilt University <i>annual departmental award for the top graduating student studying astronomy</i>	2025
Highest Honors in Astronomy, Vanderbilt University <i>for distinction in the undergraduate honors thesis and its defense</i>	2025
International Full-ride Student Grant, Vanderbilt University <i>full four-year scholarship covering all costs of undergraduate study</i>	2021-2025
Dean's List, Vanderbilt University <i>eight semesters of academic distinction</i>	2021-2025
Buchanan Fellowship, Vanderbilt University <i>supported a data-science project applying NLP to British periodical collections</i>	2022
Silver Medal, International Olympiad on Astronomy and Astrophysics <i>represented Bangladesh at IOAA 2019 and the Global e-Competition (GeCAA) 2020</i>	2020

PUBLICATION SUMMARY

13 papers, 7 as first author, 7 published or accepted in refereed journals.

108 citations from 104 citing papers ([NASA ADS Library](#)).

TELESCOPE TIME ALLOCATIONS

Principal Investigator

Large Binocular Telescope, PEPSI, 20 hours · PANTERA: ultra metal-rich stars	2026B
Large Binocular Telescope, PEPSI, 20 hours · PASTA: 36 planet-hosting stars	2026A
Large Binocular Telescope, PEPSI, 28 hours · PASTA: 43 planet-hosting stars	2025B

Co-Investigator

WIYN, NEID, NN-EXPLORE Key Program · 65 HWO target stars (PI: M. C. Johnson, submitted)	2026
Automated Planet Finder, Levy, 40 hours · metal-rich stars (PI: D. M. Rowan)	2026B
Chandra X-ray Observatory, Cycle 26 · Pink Mergers (PI: Prof. Adi Foord)	2024

INVITED TALKS AND CONFERENCE PRESENTATIONS

Monday Science Seminar, University of Wisconsin–Madison <i>Title to be announced</i>	2026
LLM Agents Seminar, NASA AI/ML STIG <i>Hands-on session II: agentic coding and research tools · recording · slides</i>	2026
AAS 245 Winter Meeting, American Astronomical Society <i>An Alternative to the Supermassive Black Hole Binary Hypothesis for J1430 (Tick Tock)</i>	2025
Cool Stars 22, Cambridge <i>Using Low Resolution Optical Spectra to Identify Young Stars and Their Properties</i>	2024
IEEE Quantum Computing and Engineering (QCE) <i>Developing Workforce in Quantum Industry: The Wond’ry Quantum Studio</i>	2023

TEACHING & MENTORING

Teaching and grading in astronomy, statistics, data science, and programming, alongside mentoring olympiad students in Bangladesh and graduate-school applicants from underrepresented backgrounds.

Mentor, Project SHORT <i>free admissions mentoring for underrepresented applicants to graduate and medical programs</i>	2025
Mentor, Bangladesh Olympiad on Astronomy and Astrophysics <i>national competition that selects and trains Bangladesh’s team for the international olympiad (IOAA)</i>	2021-2024
TA, DS 2100: Statistics for Data Science, Vanderbilt University	Spring 2025
TA, ASTR 1010: Introductory Astronomy, Vanderbilt University	Spring 2024
Grader, DS 1100: Introduction to Data Science, Vanderbilt University	Fall 2023
Instructor, Quantum Computing for Everyone, The Wond’ry Vanderbilt	Fall 2023
Grader, CS 1100: Introduction to Problem Solving with Python, Vanderbilt University	2022-2023
Tutor, Vanderbilt Student Athletics	2022

OTHER EXPERIENCES

Building an educational AI initiative, communicating new research through writing, and several years training Bangladesh’s astronomy olympiad team.

Co-founder, PorAI <i>educational AI initiative bringing personalized, tutor-like instruction to students and schools</i>	2026-Present
Author and Editor, Astrobites	2026-Present
Journal Referee (A&A)	2026-
Observer, Large Binocular Telescope, 5 nights	2026
Academic Team Member, Bangladesh Olympiad on Astronomy and Astrophysics	2021-2024
Co-lead, The Wond’ry Quantum Studio <i>student-led quantum computing program at Vanderbilt’s innovation center</i>	2022-2023
Team Leader, International Olympiad on Astronomy and Astrophysics <i>led Bangladesh’s national team and served as a question moderator at the international round</i>	2022

FIRST AUTHOR PUBLICATIONS AND PREPRINTS

7. **Serat M. Saad** & Yuan-Sen Ting (2026)
“More than Half of Recent Astronomy Papers are Written with Language-Model Assistance”
[arXiv:2609.10664](#)
6. **Serat M. Saad** & Yuan-Sen Ting (2026)
“Spectroscopic Binary Detection as Agent-Callable Tools: Detecting 40,000+ Main-Sequence Binary Candidates from SDSS DR19 APOGEE Spectra”
[The Open Journal of Astrophysics, submitted](#)
5. **Serat M. Saad**, D. M. Rowan, K. Z. Stanek, et al. (2026)
“PANTERA. I. Hunting for the Most Metal-Rich Stars in the Solar Neighborhood with High-Resolution Spectroscopy”
[The Open Journal of Astrophysics, submitted](#)
4. **Serat M. Saad** & Yuan-Sen Ting (2026)
“No Gravitational Anomaly in Wide Binaries from Forward Modeling of 3D Orbits”
[The Open Journal of Astrophysics, Vol. 9](#)

3. **Serat M. Saad** & Yuan-Sen Ting (2026)
“High-Precision Differential Radial Velocities of C3PO Wide Binaries: A Test of Modified Newtonian Dynamics (MOND)”
The Open Journal of Astrophysics, Vol. 9
2. **Serat M. Saad**, Marina Kounkel, Keivan G. Stassun, et al. (2025)
“ABYSS III: Observing Accretion Activity in Young Stars through Empirical Veiling Measurements”
The Astronomical Journal, Vol. 170, 87
1. **Serat M. Saad**, Kaitlyn Lane, Marina Kounkel, Keivan G. Stassun, et al. (2024)
“ABYSS II: Identification of Young Stars in Optical SDSS Spectra and Their Properties”
The Astronomical Journal, Vol. 167, 125

CO-AUTHOR PUBLICATIONS AND PREPRINTS

6. Yuan-Sen Ting & **Serat M. Saad** (2026)
“Armillary: A Label-Free Coordinate System for Stellar Spectra”
arXiv preprint
5. Qinghui Sun, **Serat M. Saad**, Yuan-Sen Ting, et al. (2026)
“Planets Around Solar Twins/Analog (PASTA) III: Chemical Clock Relations in Planet-Hosting Solar Analogs”
Astronomy & Astrophysics, accepted
4. SDSS Collaboration, including **Serat M. Saad** (2026)
“The Twentieth Data Release of the Sloan Digital Sky Surveys”
The Astrophysical Journal Supplement Series, submitted
3. Zimo Cheng, Sharon X. Wang, Fan Liu, . . . **Serat M. Saad**, et al. (2026)
“A Bayesian Search for Planet Engulfment Signatures in Solar Analogs”
AAS Journals, submitted
2. SDSS Collaboration, including **Serat M. Saad** (2026)
“The Nineteenth Data Release of the Sloan Digital Sky Survey”
The Astrophysical Journal Supplement Series, Vol. 285, 9
1. Yuan-Sen Ting, **Serat M. Saad**, Fan Liu, Yuting Shen (2026)
“Egent: An Autonomous Agent for Equivalent Width Measurement”
The Open Journal of Astrophysics, Vol. 9

NON-REFEREED PUBLICATIONS

6. **Serat M. Saad** (2026)
“‘Total Eclipse of the Heart’ in Astronomy”
Astrobites, August
5. **Serat M. Saad** (2026)
“Where Have All the Star Couples Gone?”
Astrobites, August
4. **Serat M. Saad** (2026)
“Cosmic Cannibalism: When Stars Eat Their Planets”
Astrobites, May
3. **Serat M. Saad** (2026)
“Wormholes Might Be More Real Than We Thought”
Astrobites, April
2. **Serat M. Saad** (2026)
“A Wandering Supermassive Black Hole Eating a Star”
Astrobites, March
1. **Serat M. Saad**
“Pixels and Persuasion”
As Seen on TV, digital exhibition, Vanderbilt University Libraries

REFERENCES

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| Krzysztof (Kris) Stanek, The Ohio State University
<i>University Distinguished Scholar; Guggenheim Fellow (2022)</i> | Ph.D. Advisor, stanek.32@osu.edu |
| Yuan-Sen Ting, The Ohio State University
<i>Inaugural Chair, NASA AI/ML Science and Technology Interest Group; Kavli Scholar (2026)</i> | Ph.D. Advisor, ting.74@osu.edu |
| Keivan G. Stassun, Vanderbilt University
<i>National Medal of Science (2024); MacArthur Fellow (2024)</i> | Undergraduate Advisor, keivan.stassun@vanderbilt.edu |